

A disproof of Conjecture 00000000502

The regularity of powers of the edge ideal of K_2 has slope 2

1 The conjecture

Conjecture 00000000502 reads:

Definition: For the edge ideal $I(G)$ of a graph G , the regularity of powers $\text{reg } I(G)^t$ is eventually linear in t , with slope denoted $a(G)$. *Conjecture:* For every bipartite graph G , $a(G) = \max\{\nu(G) - 1, \lceil (m(G) + 1)/3 \rceil\}$, where $\nu(G)$ is the matching number and $m(G)$ the induced matching number; moreover, for chordal bipartite graphs the formula simplifies to $a(G) = \nu(G) - 1$.

Reading. G is a finite simple graph on a vertex set V , $S = k[x_v : v \in V]$ is the standard graded polynomial ring over a field k , $I(G) = (x_i x_j : ij \in E(G))$, and reg is the Castelnuovo–Mumford regularity of the graded S -module $I(G)^t$. By definition $a(G)$ is the *slope* of the eventually linear function $t \mapsto \text{reg } I(G)^t$, i.e. the number a with $\text{reg } I(G)^t = at + b$ for all large t . The conjecture has two parts: (1) for every bipartite G (with $I(G) \neq 0$), $a(G) = \max\{\nu(G) - 1, \lceil (m(G) + 1)/3 \rceil\}$; (2) for every chordal bipartite G (with $I(G) \neq 0$), $a(G) = \nu(G) - 1$. Restricting to $I(G) \neq 0$ only weakens the statement. We refute (1) and (2) separately, over every field k , with the single graph $G = K_2$.

2 Definitions and sources

Definition 1 (Regularity). Let M be a finitely generated graded S -module with minimal graded free resolution $\cdots \rightarrow F_i \rightarrow \cdots \rightarrow F_0 \rightarrow M \rightarrow 0$, $F_i = \bigoplus_b S(-a_{i,b})$. Then $\text{reg } M = \max_{i,b} (a_{i,b} - i) = \max\{j - i : \beta_{i,j}(M) \neq 0\}$. Wikipedia (https://en.wikipedia.org/wiki/Castelnuovo%E2%80%93Mumford_regularity): “ R is a polynomial ring over a field k and M is a finitely generated graded R -module. Suppose M has a minimal graded free resolution [...] let b_j be the maximum of the degrees of the generators of F_j . If r is an integer such that $b_j - j \leq r$ for all j , then M is said to be r -regular. The regularity of M is the smallest such r .”

For context only (not used): Bruns–Conca–Varbaro (arXiv:2107.14734) recall “a classical result due to Cutkosky, Herzog and Trung and, independently, to Kodiyalam asserting that the regularity of powers of an homogeneous ideal I of R is eventually a linear function”. For K_2 we compute $\text{reg } I^t$ exactly, so this theorem is not needed.

Definition 2 (Matchings). A *matching* of G is a set of pairwise vertex-disjoint edges; $\nu(G)$ is the largest size of a matching. An *induced matching* is a matching M such that the subgraph of G induced on the vertices of M has edge set exactly M ; $m(G)$ is the largest size of an induced matching. A graph is *chordal bipartite* if it is bipartite and every cycle of length at least 6 has a chord.

3 The counterexample

Let $G = K_2$: vertices $0, 1$ and the single edge $\{0, 1\}$. Then $S = k[x_0, x_1]$ and $I(K_2) = (x_0x_1)$.

Lemma 3. K_2 is bipartite and chordal bipartite, $I(K_2) \neq 0$, and $\nu(K_2) = m(K_2) = 1$.

Proof. The colouring $0 \mapsto 0, 1 \mapsto 1$ is proper. A cycle has at least 3 distinct vertices, and K_2 has only 2, so K_2 has no cycles and the chord condition is vacuous. The whole graph is a matching, and it is induced, with one edge; no set of edges has more than one element since K_2 has one edge. $\square$

Lemma 4. For every $t \geq 0$, $\text{reg } I(K_2)^t = 2t$.

Proof. $I(K_2)^t = (g)$ with $g = (x_0x_1)^t \neq 0$ homogeneous of degree $2t$. *Upper bound.* $0 \rightarrow S(-2t) \rightarrow (g) \rightarrow 0, 1 \mapsto g$, is exact because S is a domain; it has no differentials, so it is minimal, and its only shift gives $a_{0,1} - 0 = 2t$. *Lower bound.* Let F be any graded free resolution of (g) , with $F_0 \rightarrow (g)$ sending the basis vector e_b (of degree $a_{0,b}$) to a homogeneous $g_b \in (g)$ of degree $a_{0,b}$. If $g_b \neq 0$, then $g \mid g_b$, so $2t = \deg g \leq \deg g_b = a_{0,b}$. If every $a_{0,b} < 2t$, all $g_b = 0$ and $(g) = (g_b)_b = 0$, a contradiction. So some $a_{0,b} \geq 2t$ and $\max(a_{i,b} - i) \geq 2t$. $\square$

Theorem 5. Over every field k : $a(K_2) = 2$, whereas $\max\{\nu(K_2) - 1, \lceil (m(K_2) + 1)/3 \rceil\} = \max\{0, 1\} = 1$ and $\nu(K_2) - 1 = 0$. Hence both parts (1) and (2) of Conjecture 00000000502 are false.

Proof. By Lemma 4, $\text{reg } I(K_2)^t = 2t$ for all t , so it is eventually linear with slope 2. The slope is unique: if $2t = at + b$ for all $t \geq N$, then comparing $t = N$ and $t = N + 1$ gives $a = 2$. By Lemma 3, $\nu = m = 1$, so $\lceil 2/3 \rceil = 1$ and the two formulas give 1 and 0, neither equal to 2. Since K_2 is bipartite and chordal bipartite with $I(K_2) \neq 0$, it violates both parts. $\square$

Remark 6 (Other readings). (a) If $a(G)$ were misread as the constant term b in $\text{reg } I(G)^t = 2t + b$, part (1) still fails: $b(K_2) = 0 \neq 1$ (Lean: `constant_term_reading_false`). Under that misreading part (2) would hold for K_2 ($0 = \nu - 1$), so for (2) only the literal reading is refuted. (b) Using $\text{reg}(S/I^t) = \text{reg}(I^t) - 1$ instead of $\text{reg } I^t$ does not change the slope. (c) K_2 is the smallest example. Under the literal reading every graph with an edge has slope 2, because $I(G)$ is generated in degree 2, so the formula fails whenever its value is not 2. Only the K_2 case is used here.

4 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). `Conjecture502.lean` imports the single source file `Conjecture502/Basic.lean` (344 lines, namespace C502).

Definitions.

- `edgeIdeal k G := Ideal.span {p | exists i j, G.Adj i j and p = X i * X j}` in `MvPolynomial V k`.
- `HomogDeg p d`: $p = 0$ or p is homogeneous (`IsHomogeneous`) of degree $n \in \mathbb{N}$ with $n = d$.
- `GradedFreeRes J`: ranks `rank i`, shifts `shift i b : Int`, the images `gen b` of the basis of F_0 , matrices `d i` of $F_{i+1} \rightarrow F_i$ with entries homogeneous of degree `shift (i+1) b - shift i a`; `span_gen`: the `gen b` generate J ; `exact_zero`, `exact_succ`: exactness at F_0 (kernel of $F_0 \rightarrow J = \text{image of } d_0$) and at every F_{i+1} ; `finite_length`. `IsMinimal`: every matrix entry has constant coefficient 0, i.e. lies in (x_v) .

- `reg J := sInf {r | exists F : GradedFreeRes J, F.IsMinimal and forall i b, F.shift i b - i <= r}`: the smallest r with $b_j - j \leq r$ for all j in a minimal graded free resolution, exactly as in Definition 1. All minimal resolutions have the same shifts, and taking the infimum means no particular one has to be chosen.
- `matchingNumber G, inducedMatchingNumber G`: `sSup` of `M.edgeSet.ncard` over Mathlib sub-graphs `M` with `M.IsMatching` (and `M.IsInduced`).
- `Bipartite` is Mathlib's `SimpleGraph.IsBipartite`. `IsChordalBipartite G`: bipartite and every `c : G.Walk u u` with `c.IsCycle` and `6 <= c.length` has vertices x, y in its support with `G.Adj x y` and `s(x,y)` not an edge of `c`.
- `HasEventualSlope f a := exists b : Real, eventually (atTop) f t = a * t + b`.
- `GeneralFormula k`: for all `V : Type, [Fintype V], G`, if `G.IsBipartite` and `edgeIdeal k G != bot`, then `HasEventualSlope (fun t => reg (edgeIdeal k G ^ t)) (max (nu G - 1) (ceil ((m G + 1)/3)))`. `ChordalFormula k`: the same for `IsChordalBipartite G` with slope `nu G - 1`. `Conjecture k := GeneralFormula k and ChordalFormula k`.

Theorems. `reg_K2_pow : reg (edgeIdeal k K2 ^ t) = 2 * t` (Lemma 4; resolution res, lower bound shift_lower_bound); `matchingNumber_K2, inducedMatchingNumber_K2, K2_isBipartite, K2_isChordalBipartite, edgeIdeal_K2_ne_bot` (Lemma 3); `slope_K2 : HasEventualSlope (...) a <-> a = 2`; `generalFormula_false, chordalFormula_false; constant_term_reading_false`; and the main theorem

`conjecture_00000000502_false (k : Type) [Field k] : ¬GeneralFormula k ∧ ¬ChordalFormula k ∧ ¬Conjecture k`.

Axiom audit. There is no sorry, no `native_decide` and no added axiom. The file compiles with no warnings.

`'C502.conjecture_00000000502_false'` depends on axioms:
`[propext, Classical.choice, Quot.sound]`
`'C502.constant_term_reading_false'` depends on axioms:
`[propext, Classical.choice, Quot.sound]`

To check: `cd lean; lake exe cache get; lake build`.